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Effects of Herbal Preparation on Libido and Semen Quality in Boars

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Contents

The objective of this study was to investigate the effects of a preparation from herbal extracts (PHE) on libido and semen quality in breeding artificial insemination boars. Ten fertile boars were divided into control and experimental groups according to significant difference of libido. There were no differences in semen quality between groups. Animals were fed a commercial feeding mixture for boars. The feeding mixture for the experimental group was enriched with PHE, which was prepared from *Eurycoma longifolia*, *Tribulus terrestris* and *Leuzea carthamoides*. Duration of the experiment was 10 weeks. Samples of ejaculate were collected weekly. Libido was evaluated according to a scale of 0–5 points. Semen volume, sperm motility, percentage of viable spermatozoa, sperm concentration, morphologically abnormal spermatozoa, daily sperm production and sperm survival were assessed. Amounts of mineral components and free amino acids were analysed in seminal plasma. Significant differences were found in these parameters: libido (4.05 ± 0.22 vs 3.48 ± 0.78; $p < 0.001$), semen volume (331.75 ± 61.91 vs 263.13 ± 87.17 g; $p < 0.001$), sperm concentration (386.25 ± 107.95 vs $487.25 \pm 165.50 \times 10^3/mm^3$; $p < 0.01$), morphologically abnormal spermatozoa (15.94 ± 11.08 vs $20.88 \pm 9.19\%$; $p < 0.001$) and Mg concentration (28.36 ± 11.59 vs 20.27 ± 13.93 mM; $p < 0.05$). The experimental group's libido was increased by 20% in comparison with the beginning of the experiment. Results of this study showed positive effect of PHE on libido and some parameters of boar semen quality.

Introduction

Breeding boars whose semen is collected for artificial insemination (AI) make up a relatively small proportion of the pig population. At present, the number of breeding boars is decreasing at AI centres in the Czech Republic. This means that each boar must have high reproductive output and good breeding results. Reproductive output includes evaluations of libido, mating ability and semen quality. Therefore, it remains pertinent to find new ways of maintaining and improving boar reproductive ability. One possible approach is to use plant extracts to stimulate the animals' reproductive system.

Roots of *Eurycoma longifolia* (fam. Simaroubaceae), a plant native to Indonesia and Malaysia, contain quassinoids with anabolic, antimalarial, cytostatic and aphrodisiac activity; squalene derivatives; biphenylneolignans; tirucallane-type triterpenes; canthine; and β-carboline alkaloids (Ang et al. 2000). The effects of various fractions from *E. longifolia* root extract on the initiation of sexual performance in rats were studied by Ang and Lee (2002), Ang and Sim (1997), Ang et al. (2003). The studied fractions of the extract enhanced the sexual qualities by improving libido, modifying orientation activities and decreasing the rats' hesitation time.

The herb *Tribulus terrestris* (fam. Zygophyllaceae), a plant also known as puncture vine and native to the Mediterranean region, contains a wide range of biologically active substances, including steroidal saponins, flavonoids, alkaloids, unsaturated fatty acids, vitamins and tannins (Adaikan et al. 2000). The mixture of these substances has aphrodisiac effects, improves spermatogenesis and decreases manifestation of erectile impotence (Oplétal et al. 2008). Animal studies with rats, rabbits and primates have demonstrated that the administration of *T. terrestris* extract can evoke statistically significant increases in levels of testosterone, dihydrotestosterone and dehydroepiandrosterone (Gauthaman and Ganesan 2008) and induces effects suggesting aphrodisiac activity (Gauthaman et al. 2002).

Leuzea carthamoides (syn. *Rhaponticum carthamoides*) is an herb of Siberian origin belonging to the family Asteraceae. The main active constituents of *L. carthamoides* are ecdysteroids (5 β-cholest-6-on-7-ene derivatives) and flavonoids (Varga et al. 1990). The component 20-hydroxyecdysone plays an important role in the biological activity of this plant (Syrov and Kurmukov 1976). Koleckar et al. (2008) observed an antioxidant effect of *L. carthamoides* extract. The extract has adaptogenic effects, decreases unfavourable influence of various stress factors, increases proteosynthesis and enhances libido (Oplétal 2010).

We previously published a review concerning natural compounds potentially influencing pig reproduction (Oplétal et al. 2008), and on the basis of this information, these specific herbs were chosen for our experiment. An herbal preparation (PHE) was prepared using *E. longifolia*, *T. terrestris* and *L. carthamoides*. The combination of these three herbal extracts should evoke aphrodisiac effect, including enhancing libido, while improving spermatogenesis and erectile ability.

The objective of this study was to investigate the effects of the PHE on libido and semen quality of breeding boars at an AI centre.

Materials and Methods

Ten fertile hybrid boars of various ages (1–3 years) coming from one AI centre in the Czech Republic were used in this study. The experiment was conducted during summer. All boars were kept under the same housing, feeding and breeding conditions. Boars were divided into two groups according to the significantly different ($p < 0.05$) libido (for libido classification see Table 1): control group C ($n = 5$, libido = 4 points) and experimental group E ($n = 5$, libido = 3 points). Animals were fed a commercial feeding mixture for

Table 1. Libido classification of boars at insemination station

Point scale	Evaluation	Characterization of libido
0	No jump	No sexual interest and unable to collect
1	Inconvenient	Very little sexual interest and slow to achieve erection and ejaculation
2	Substandard	Little sexual interest, several times follow out the jump, longer time for erection and ejaculation
3	Standard	Moderate sexual interest, using incentive to jump with consecutive erection and ejaculation
4	Very good	Great sexual interest, longer search reflex with consecutive erection and ejaculation
5	Excellent	Intense sexual interest, immediate erection and ejaculation

breeding boars. The PHE was added to the feeding mixture for the E group daily. Duration of the experiment was 10 weeks. During the first and the last week (control period), group E was fed the feeding mixture without PHE, while from the second through ninth weeks, the feeding mixture was enriched by PHE. Ejaculates were collected by a gloved-hand technique, and libido was classified weekly.

The PHE consists of *E. longifolia* standardized extract (Eurycomae extractum siccum 50 : 1; Sumatra Pasak bumi, JL Letda Simono, Medan, Indonesia, 50% of quassinoids), *T. terrestris* standardized extract (Tribuli extractum siccum; Hunan OSST HerbPharma, Inc., Changsha, China, 40% of steroidal saponins) and *L. carthamoides* standardized extract [prepared in the laboratory of the Department of Pharmaceutical Botany and Ecology, Pharmaceutical Faculty in Hradec Králové, Charles University in Prague, 1.50% of 20-hydroxyecdysone (dry weight) and lauric acid (Merck Co., Darmstadt, Germany)]. Lauric acid masks the bitter taste and it has an antimicrobial effect. Composition of the herbal preparation is presented in Table 2.

The following semen quality parameters were evaluated in samples of ejaculate: semen volume, sperm motility, percentage of viable spermatozoa, sperm concentration, morphologically abnormal spermatozoa, daily sperm production and sperm survival. Quantities of mineral components and free amino acids were analysed in the seminal plasma.

Sperm motility was evaluated subjectively by microscopic estimation of the number of sperm moving in a visual field of phase-contrast microscopy with a heating stage (38°C) at 100× magnification. Each sample was examined at three different microscopic fields, and motility was expressed as percentage of sperm showing normal forward progressive movements (Věžník et al. 2004). Sperm concentration was estimated using a Bűrker counting chamber. Morphologically abnormal spermatozoa were evaluated according to the staining method of Čeřovský (1976) and evaluated microscopically under oil immersion and 1500× magnification. Daily sperm production was calculated from total sperm output per ejaculate and length of the interval since previous collection. Percentage of viable spermatozoa was estimated by supravital staining technique using the eosin–nigrosin stain mixture (Věžník et al. 2004). One drop from each sample was mixed with one drop of 1% eosin Y, then two drops of 10% nigrosin were added after 30 s. Two hundred spermatozoa per slide were evaluated under a light microscope (1500×). Sperm survival was assessed in diluted boar semen. The semen

Table 2. Composition of herbal preparation

Ingredient	
Eurycoma longifolia[a]	1 mg
Tribulus terrestris[b]	1 mg
Leuzea carthamoides[c]	6.66 mg
Lauric acid	9 mg

[a]With content 75% by weight quassinoids per 1 kg live weight/day.
[b]With content 40% by weight of steroid saponins per 1 kg live weight/day.
[c]Quantity of *Leuzea carthamoides* extract corresponding to 0.1 mg of 20-hydroxyecdysone per 1 kg live weight/day.

was diluted (dilution ratio 1 : 8) in a commercial boar semen extender SUS (HEMA Malšice, Malšice, Czech Republic) and was stored at a temperature of 17°C up to 96 h. Sperm motility was evaluated 24, 48, 72 and 96 h after semen dilution. Mineral components and free amino acids were determined from seminal plasma. Seminal plasma was separated from spermatozoa by centrifugation (JOUAN CR3i; JUAN Industries S.A.S, St. Herblain, France) at 492 *g* for 10 min and frozen at −20°C for later analysis. Concentrations of Na, K, Ca, Mg and Zn ions in seminal plasma were assessed by atomic absorption spectrophotometer (AAS SP9; Pye Unicam Ltd., Cambridge, UK). Free amino acids (alanine, aspartic acid, glutamic acid, glycine, isoleucine, leucine, lysine, methionine, phenylalanine, serine, taurine, threonine, tyrosine and valine) were analysed by liquid chromatography (AAA T 339 M; Mikrotechna Praha, Prague, Czech Republic). Libido was evaluated subjectively by insemination station staff according to a 5-point classification scale. Zero denotes minimum, and five denotes maximum value of libido (Table 1).

Basic statistical characteristics of the results (arithmetic means, standard deviations and significance) were calculated using the QC Expert program (TriloByte Statistical Software, s.r.o., Pardubice, Czech Republic). Statistical significance was checked using the unpaired *t*-test at significance levels of $p < 0.05$, $p < 0.01$, and $p < 0.001$.

Results

Breeding boars were divided into control and experimental groups according to the values of libido and this was the only significantly different parameter ($p < 0.05$) in the first week (control period) of the experiment. Mean values of all parameters are shown in Table 3. Mean values for libido and parameters of boar semen quality during the tested period (when feeding mixture contained PHE) are presented in Tables 4–6.

Table 3. Mean values of boar semen quality in C and E groups in the first semen collection (control period, feeding mixture without preparation from herbal extracts)

	Control group (n = 5)	Experimental group (n = 5)
Parameter	Mean ± SD	Mean ± SD
Libido	4.00 ± 0.71	3.00 ± 0.71*
Semen volume (g)	352.00 ± 34.21	276.00 ± 98.64
Sperm motility (%)	75.00 ± 5.00	76.00 ± 4.18
Viable spermatozoa (%)	70.80 ± 6.06	73.60 ± 2.70
Morphologically abnormal spermatozoa (%)	14.40 ± 7.81	17.20 ± 11.06
Sperm concentration ($mm^3 \times 10^3$)	318.00 ± 29.23	381.50 ± 98.43
Daily sperm production ($\times 10^9$)	15.96 ± 1.76	14.80 ± 6.46

*p < 0.05.

Table 4. Comparison of mean values for boar semen quality in C and E groups during tested period (when feeding mixture contained preparation from herbal extracts)

	Control group (n = 5)	Experimental group (n = 5)
Parameter	Mean ± SD	Mean ± SD
Libido	4.05 ± 0.22	3.48 ± 0.78**
Semen volume (g)	331.75 ± 61.91	263.13 ± 87.17**
Sperm motility (%)	71.13 ± 6.35	70.75 ± 6.46
Viable spermatozoa (%)	66.25 ± 4.78	67.55 ± 4.32
Morphologically abnormal spermatozoa (%)	15.94 ± 11.08	20.88 ± 9.19*
Sperm concentration ($mm^3 \times 10^3$)	386.25 ± 107.95	487.25 ± 165.50*
Daily sperm production ($\times 10^9$)	17.95 ± 4.60	17.63 ± 6.52
Sperm motility 24 h (%)	47.25 ± 6.98	47.63 ± 6.30
Sperm motility 48 h (%)	42.88 ± 7.32	41.13 ± 6.84
Sperm motility 72 h (%)	38.50 ± 8.15	36.50 ± 6.62
Sperm motility 96 h (%)	35.00 ± 8.37	34.50 ± 7.14

*p < 0.01; **p < 0.001.

Table 5. Mineral components (mM) in seminal plasma in C and E groups during tested period (when feeding mixture contained preparation from herbal extracts)

	Control group (n = 5)	Experimental group (n = 5)
Parameter	Mean ± SD	Mean ± SD
Ca	1.24 ± 0.62	1.10 ± 0.37
Zn	0.42 ± 0.11	0.42 ± 0.29
K	15.92 ± 3.36	16.94 ± 2.68
Mg	28.36 ± 11.59	20.27 ± 13.93*
Na	130.50 ± 36.59	148.55 ± 54.45

*p < 0.05.

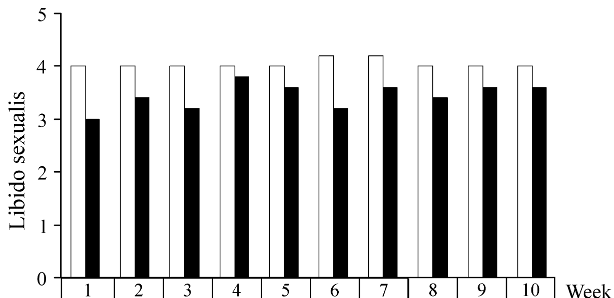
Comparison of mean values for boar semen quality between groups C and E is shown in Table 4. Significant differences were found in semen quality between the groups. Libido and semen volume were significantly higher in the C group in comparison with the E group (4.05 ± 0.22 vs 3.48 ± 0.78, $p < 0.001$; 331.75 ± 61.91 vs 263.13 ± 87.17 g, $p < 0.001$, respectively). The mean values of libido during the entire experiment are illustrated in Fig. 1. The C group had the same values for libido (4.00) during the whole experiment except for the sixth (4.20) and seventh (4.20) weeks, while the E group had higher values during the tested period (when feeding mixture contained PHE) and also in the last week (when feeding mixture was without PHE) than in the first week (control period). Libido was increased by 20% in group E (3.60 vs 3.00). The percentage of morphologically abnormal spermatozoa was significantly higher in the E group ($p < 0.001$). The most considerable effect of PHE was observed in the

Table 6. Concentrations of free amino acids (μmol/100 ml) in seminal plasma in C and E groups during tested period (when feeding mixture contained preparation from herbal extracts)

	Control group (n = 5)	Experimental group (n = 5)
Parameter	Mean ± SD	Mean ± SD
Alanine	2.88 ± 1.33	2.83 ± 1.45
Aspartic acid	2.24 ± 1.07	2.29 ± 0.92
Glutamic acid	28.10 ± 5.10	31.21 ± 10.29
Glycine	12.81 ± 3.89	12.79 ± 7.11
Isoleucine	1.21 ± 0.69	1.22 ± 0.76
Leucine	1.39 ± 0.73	1.43 ± 0.84
Lysine	0.75 ± 0.33	0.73 ± 0.39
Methionine	0.52 ± 0.41	0.42 ± 0.30
Phenylalanine	1.23 ± 0.72	1.11 ± 0.40
Serine	1.65 ± 0.66	1.47 ± 0.57
Taurine	7.77 ± 2.10	8.59 ± 3.53
Threonine	1.69 ± 0.73	1.64 ± 0.65
Tyrosine	1.27 ± 0.79	1.22 ± 0.62
Valine	2.29 ± 1.06	2.11 ± 1.20

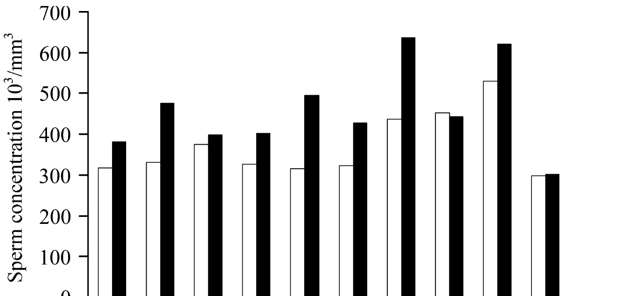
sperm concentration of E group in comparison with C group (487.25 ± 165.50 vs 386.25 ± $107.95 \times 10^3/mm^3$, $p < 0.01$). Figure 2 shows sperm concentration during the whole experiment. No differences between E and C groups were found in sperm motility, percentage of viable spermatozoa, sperm survival and daily sperm production ($p > 0.05$).

The contents of mineral components contained in seminal plasma are shown in Table 5. Significant differences in Mg ions concentration were detected between C and E groups during the tested period (20.27 ± 13.93 vs 28.36 ± 11.59 mM; $p < 0.05$). The



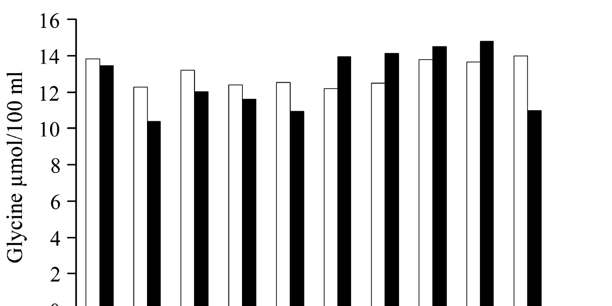
	1	2	3	4	5	6	7	8	9	10
□ C group	4.00	4.00	4.00	4.00	4.00	4.20	4.20	4.00	4.00	4.00
■ E group	3.00	3.40	3.20	3.80	3.60	3.20	3.60	3.40	3.60	3.60

Fig. 1. The mean values of libido sexualis during the whole experiment (1st and 10th week without preparation from herbal extracts in E group)



	1	2	3	4	5	6	7	8	9	10
□ C group	318.00	330.50	375.00	326.50	315.00	322.50	437.50	453.00	530.00	298.00
■ E group	381.50	475.00	399.00	402.00	494.50	426.50	637.00	442.50	621.50	301.00

Fig. 2. The mean values of sperm concentration during the whole experiment (1st and 10th week without preparation from herbal extracts in E group)



	1	2	3	4	5	6	7	8	9	10
□ C group	13.84	12.27	13.18	12.39	12.52	12.19	12.49	13.79	13.65	14.01
■ E group	13.43	10.41	12.00	11.62	10.92	13.96	14.12	14.51	14.77	10.97

Fig. 3. The mean values of glycine concentration during the whole experiment (1st and 10th week without preparation from herbal extracts in E group)

data summarized in Table 6 show the mean concentrations of free amino acids in seminal plasma. No significant differences between groups were found during the tested period. Changes in glycine concentration during the whole experiment are reported in Fig. 3.

The results from the tenth collection (control period – feeding mixture without PHE) are in Table 7. Among boar semen quality parameters, there was a significant difference between the C and E groups ($p < 0.05$) only in the percentages of morphologically abnormal spermatozoa.

Discussion

The results of our experiment demonstrate a positive effect of PHE on libido and some parameters of boar semen quality. Libido was increased by 20% in group E. This might be explained by an aphrodisiac effect of PHE. Quassinoids (*E. longifolia*), saponins (*T. terrestris*) and ecdysteroids (*L. carthamoides*) had stimulative effects on sexual activity of boars. Quassinoids are substances influencing libido by a direct brain effect; they do not intervene in other functions and specifically do not evoke side effects. They cause increase in testosterone in Leydig cells together with sexual motivation. Saponins increase the levels of dehydroepiandrosterone, luteinizing hormone (DHEA, LH) and testosterone and do not affect follicle-stimulating hormone (FSH) (Oplétal et al. 2008). Gauthaman and Ganesan (2008) noted hormonal effects of *T. terrestris* in primates, rabbits and rats; e.g. in primates, increases were observed for testosterone (+52%), dihydrotestosterone (+31%) and DHEA (+29%). Brown et al. (2001) found that *T. terrestris* increased ($p < 0.05$) free testosterone (+38%) and dihydrotestosterone (+71%). On the other hand, Milanov et al. (1981) found that FSH concentration in men (mean value was 13.38 mIU/ml both prior to and post-treatment, $p > 0.05$) was not affected by *T. terrestris*, although LH and testosterone (mean values, respectively, were 13.0 prior to and 37.25 mIU/ml post-treatment, and 628 prior to and 882 ng % post-treatment, $p < 0.05$) were increased. Ecdysteroids also increase libido sexualis but their main role is increases proteosynthesis and in adaptogenic effects (Koleckar et al. 2008). Ang and Sim (1997) and Gauthaman and Ganesan (2008) have described positive effects of *T. terrestris* and *E. longifolia* extracts on libido in male rats. Kistanova et al. (2005) had found that extract from *T. terrestris* manifested a good libido and active sexual behaviour in rams.

In our study, we determined that an 8-week period from the very beginning of feeding PHE is necessary for improvements in semen quality to appear because spermatogenesis takes approximately 40 days, and the sperm transit through the epididymis takes approximately 10 days in the boar (Franca et al. 2005). A positive effect of PHE was detected on sperm concentration in the E group in comparison with the C group ($p < 0.01$) during the test period. The greatest increase in sperm concentration was noted in the seventh and ninth weeks in the E group. It is likely that the increase in sperm concentration is related to the influence of *T. terrestris* extract. Kistanova et al. (2005) had shown that the same plant improved spermatozoa number and also the time of viability and motility of ram sperm. Viktorov et al. (1994) demonstrated a positive effect of *T. terrestris* on spermatogenesis in mice. The increased sperm concentration is a very valuable result, because it is known that in summer, sperm concentration declines because of heat stress.

Table 7. Comparison of mean values of boar semen quality in C and E groups – the 10th semen collection (control period, feeding mixture without preparation from herbal extracts)

Parameter	Control group (n = 5) Mean ± SD	Experimental group (n = 5) Mean ± SD
Libido	4.00 ± 0.00	3.60 ± 0.89
Semen volume (g)	336.00 ± 39.27	281.00 ± 91.13
Sperm motility (%)	72.00 ± 7.58	69.00 ± 7.42
Viable spermatozoa (%)	66.40 ± 3.29	66.80 ± 5.07
Morphologically abnormal spermatozoa (%)	9.80 ± 4.59	25.40 ± 9.74*
Sperm concentration ($mm^3 \times 10^3$)	298.00 ± 87.06	301.00 ± 109.60
Daily sperm production ($\times 10^9$)	13.98 ± 2.57	11.56 ± 4.23

*$p < 0.05$.

On the other hand, the rate of morphologically abnormal spermatozoa was affected by feeding PHE to the E group. The percentage of abnormal spermatozoa was increased by 31% ($p < 0.01$) in the E group compared to the C group during the tested period. The percentage of morphologically abnormal spermatozoa did not exceed 25% in the E group, which is tolerable for AI in the Czech Republic. Moreover, we observed lower semen volume in the E group in comparison with the C group ($p < 0.001$) during the tested period. The values of sperm motility, percentage of viable spermatozoa, daily sperm production and sperm survival were not affected by PHE ($p > 0.05$).

The only significant difference ($p < 0.05$) in mineral content was found for Mg ions. Concentration of Mg ions was lower in the E than C group. The contents of other mineral components analysed in seminal plasma (Ca, Zn, K and Na) were not significantly different ($p > 0.05$). Máchal et al. (2007) had discovered a negative correlation between sperm quality and concentration of Mg in both blood and seminal plasma. We also observed negative correlation ($r = -0.13$, $p < 0.05$) between the amount of Mg ions and sperm concentration in the C group.

The content of free amino acids in seminal plasma was not significantly influenced by PHE in the E group during the tested period ($p > 0.05$). Glutamic acid and glycine were determined to be the predominant free amino acids in seminal plasma. Johnson et al. (1972) had reported the predominant rate of these amino acids and also that glutamic acid, hypotaurine and taurine were in highest concentrations in plasma from the highest sperm-containing fractions derivable from the epididymal plasma, while glycine was predominant in sperm-poor plasma in the seminal vesicle fluid. The function of seminal plasma free amino acids is not wholly established although they have been shown to act as fuels for the spermatozoa, to create favourable conditions for cell survival and to be probably involved in detoxification functions (Ibrahim and Boldizsár 1981). A significantly negative relationship between morphologically abnormal spermatozoa and the concentration of free amino acids in seminal plasma was found by Čeřovský et al. (2007).

In conclusion, the addition of PHE had a positive effect on libido and some parameters of boar semen quality of breeding boars at an AI centre. The PHE especially improved libido, increased sperm concentration and positively influenced concentration of Mg ions. On the other hand, it was found a negative effect of PHE on the rate of morphologically abnormal spermatozoa. This experiment suggests the possibility for utilizing PHE containing *E. longifolia*, *T. terrestris* and *L. carthamoides* in commercial feeding mixtures for breeding AI boars, and especially for improving libido.

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Conflict of interest

None of the authors have any conflict of interest to declare.

Author contributions

Frydrychova Sona (designed study, analysed data, drafted paper), Opletal Lubomir (designed study, drafted paper), Lustykova Alena (analysed data), Rozkot Miroslav (designed study, drafted paper), Lipensky Jan (analysed data), Macáková Kateřina (corrected for better English language).

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